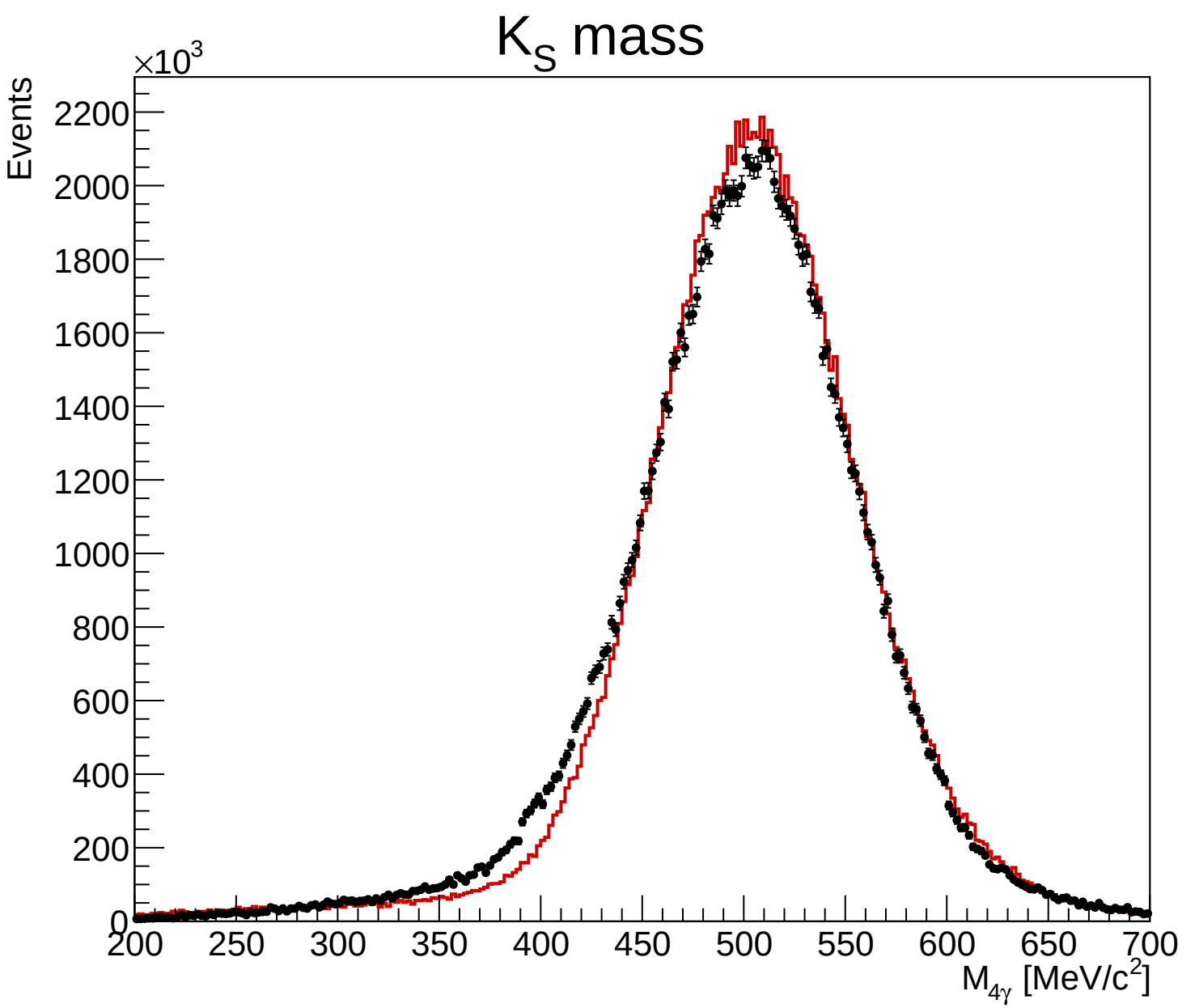
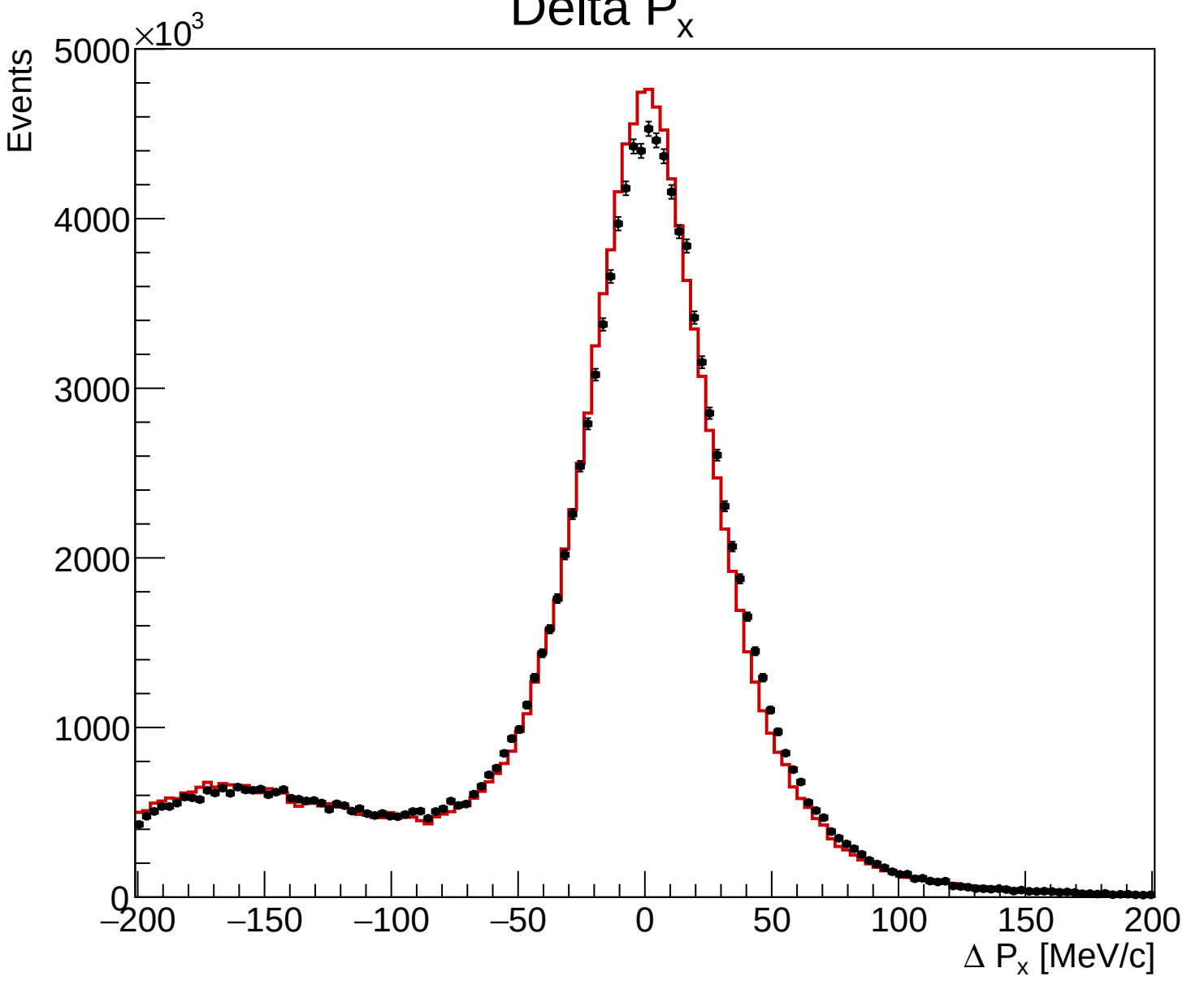
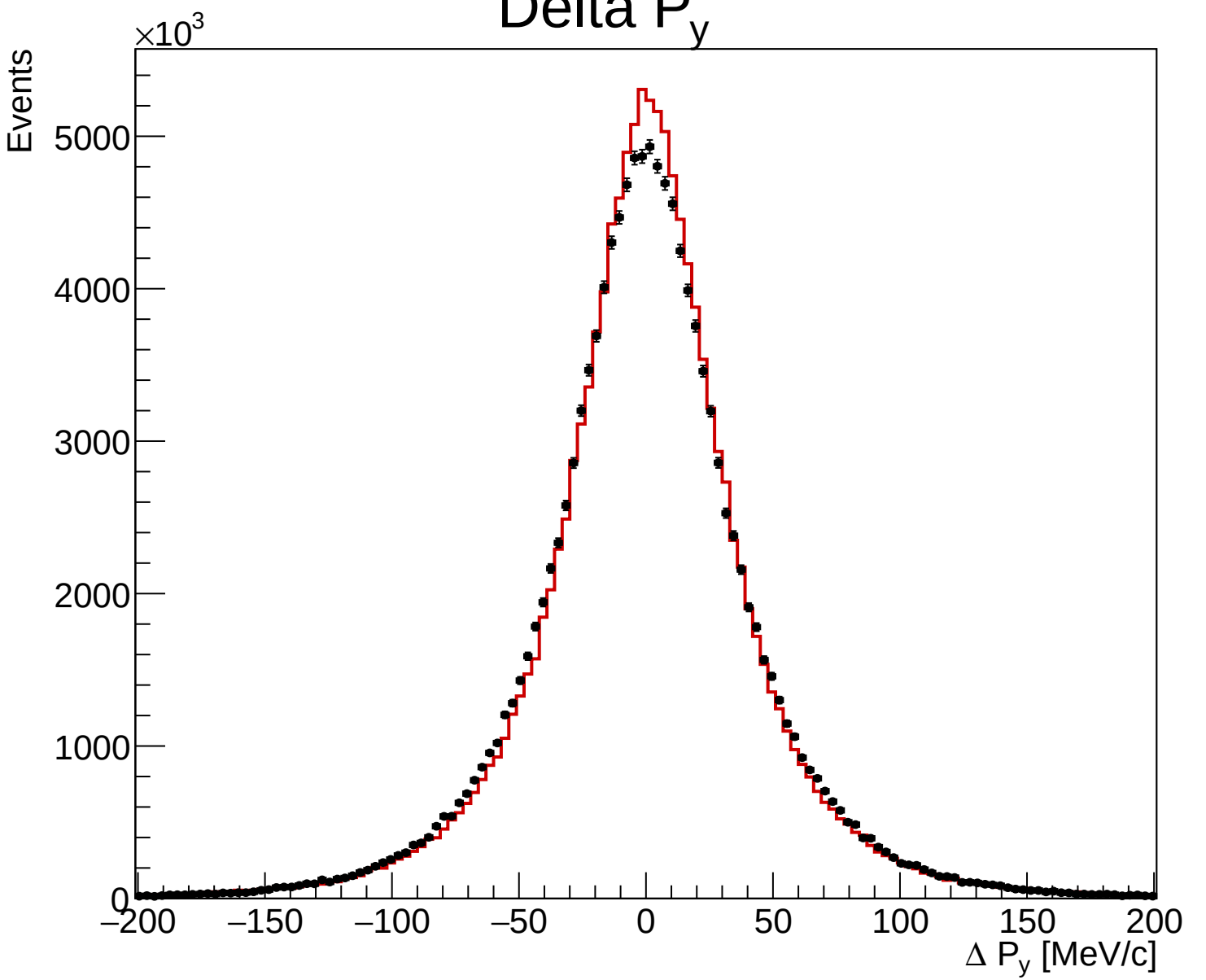
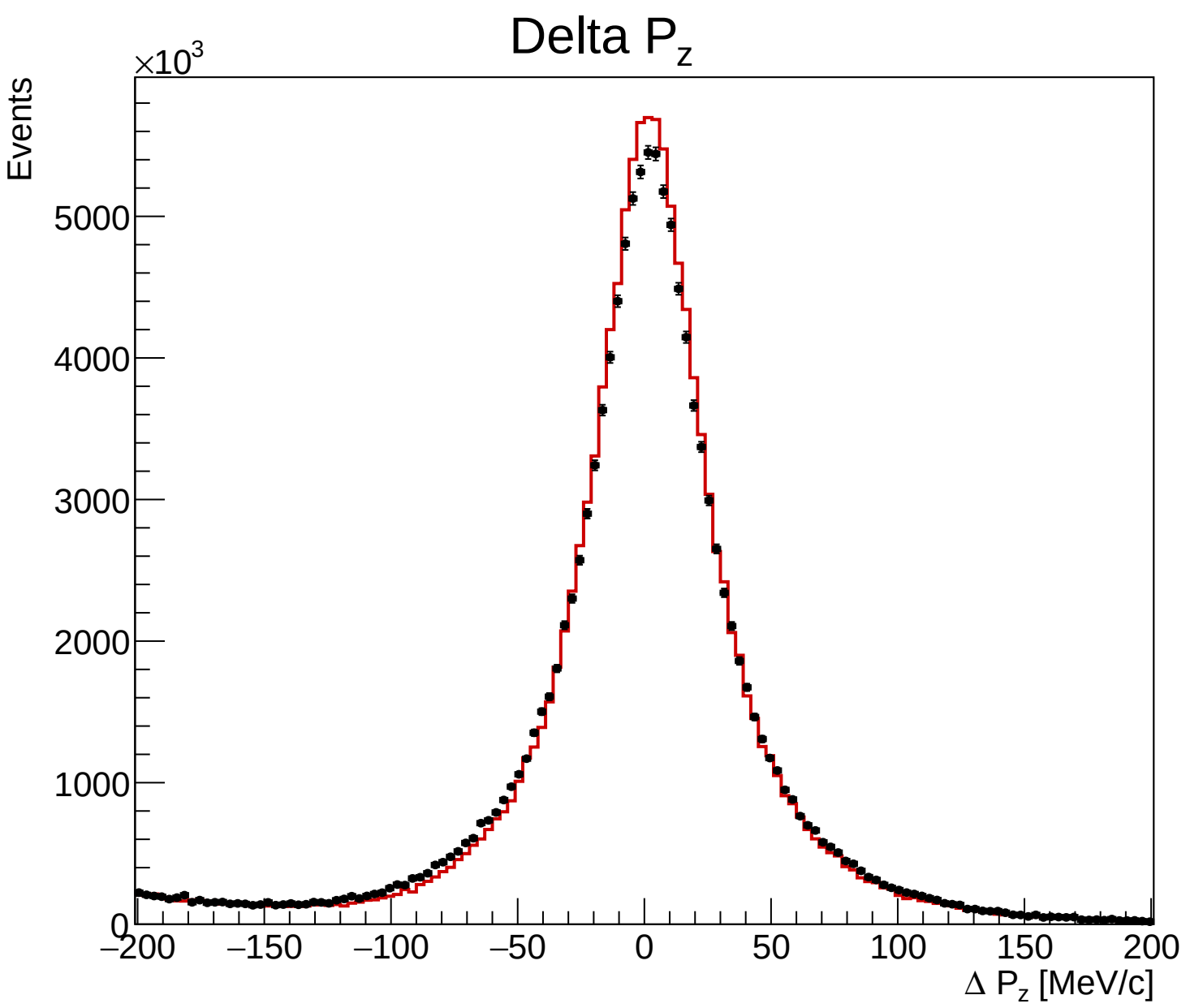
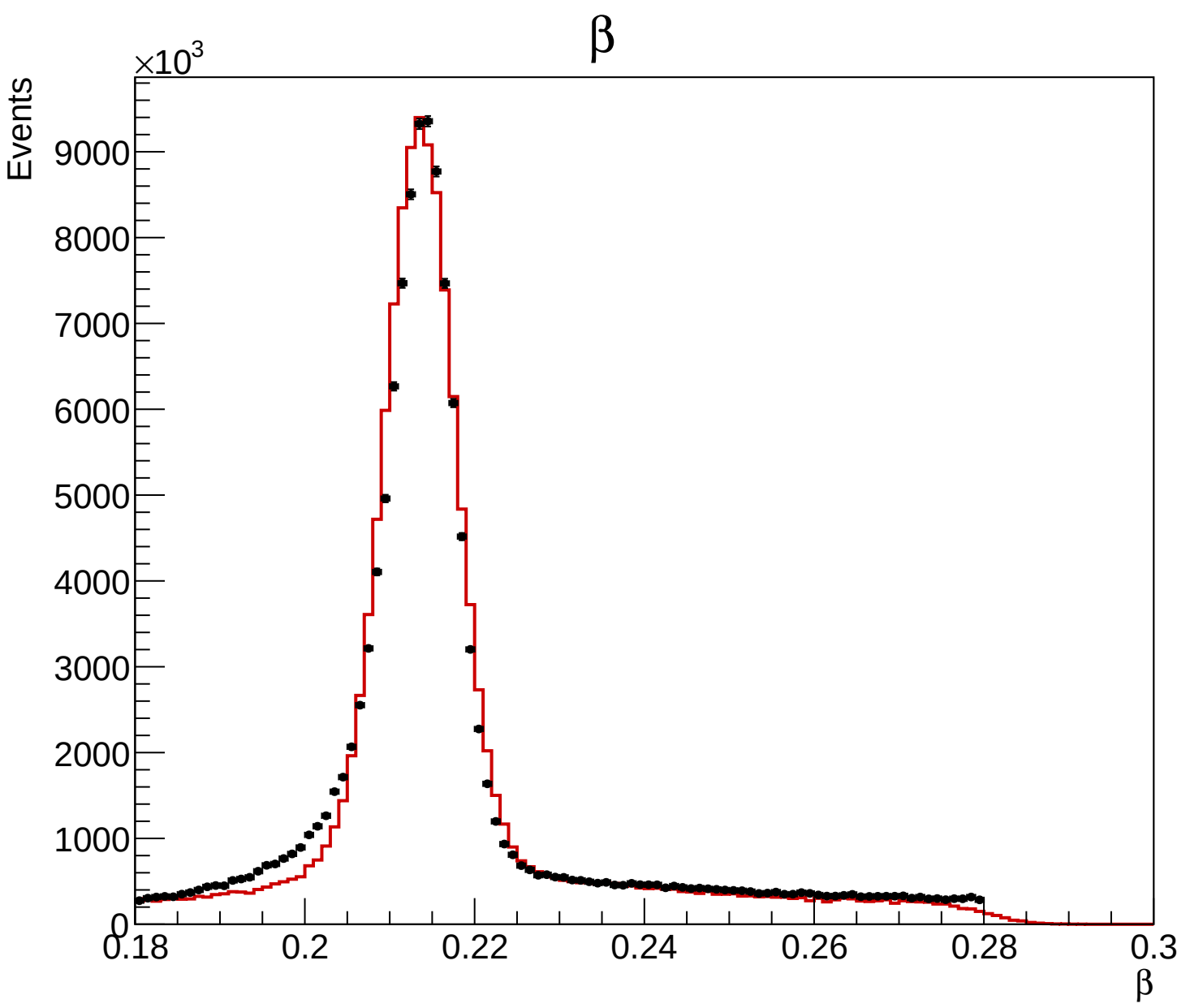
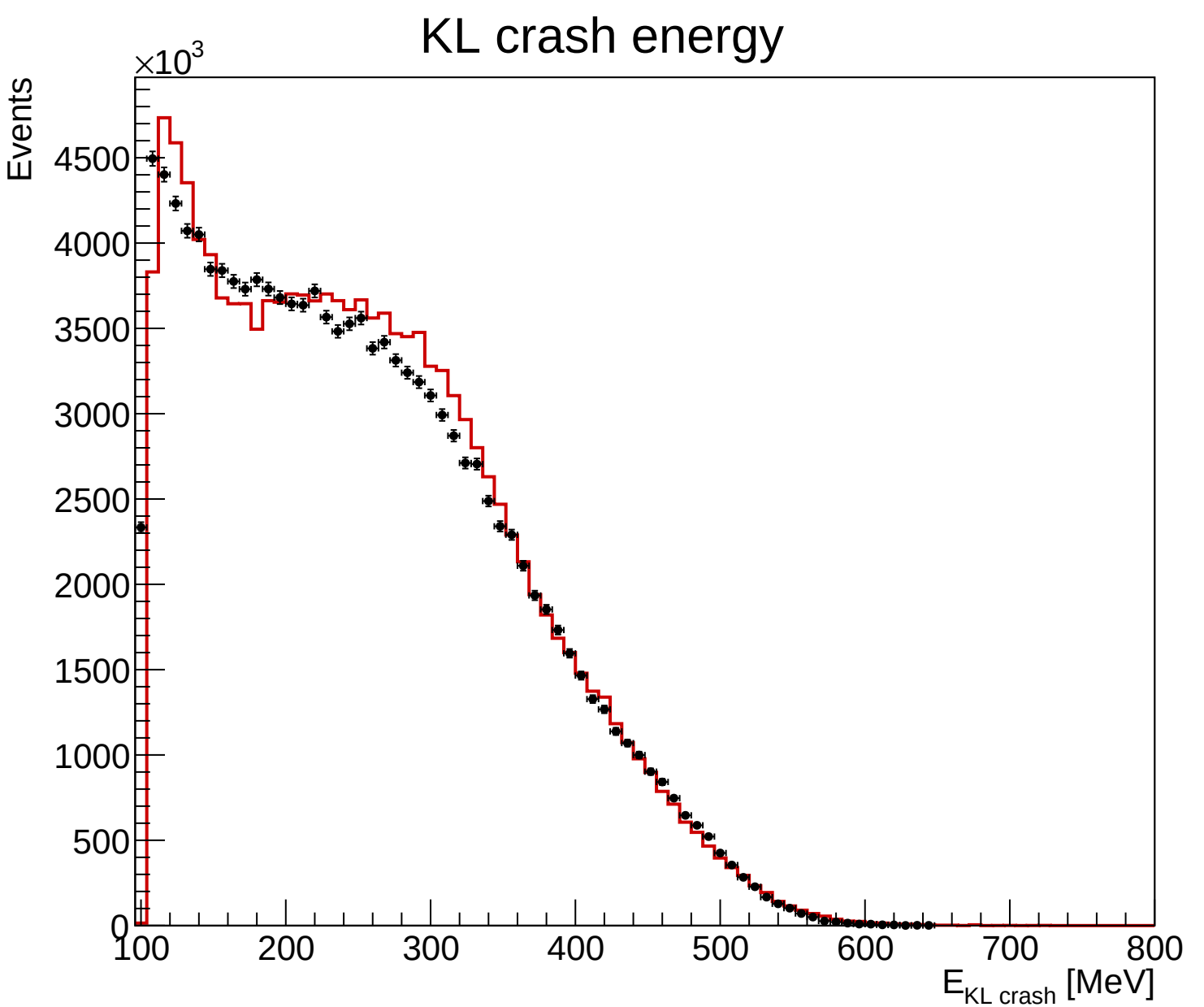

K_S mass
$\times 10^3$
Events
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
200
250
300
350
400
450
500
550
600
650
700
$M_{4\gamma}$ [MeV/c^2]

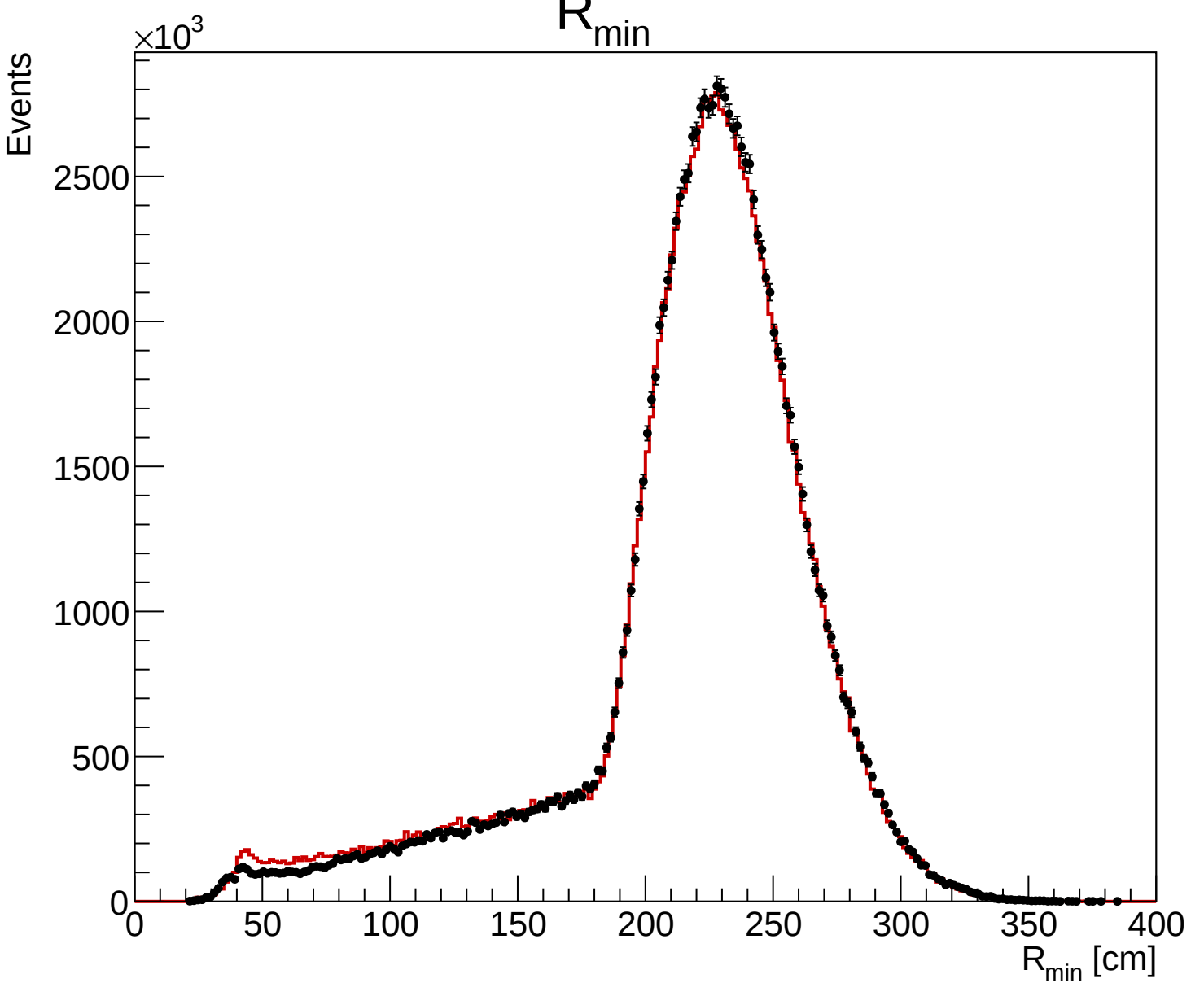

Delta P_x
$\times 10^3$
Events
5000
4000
3000
2000
1000
0
–200
–150
–100
–50
0
50
100
150
200
ΔP_x [MeV/c]

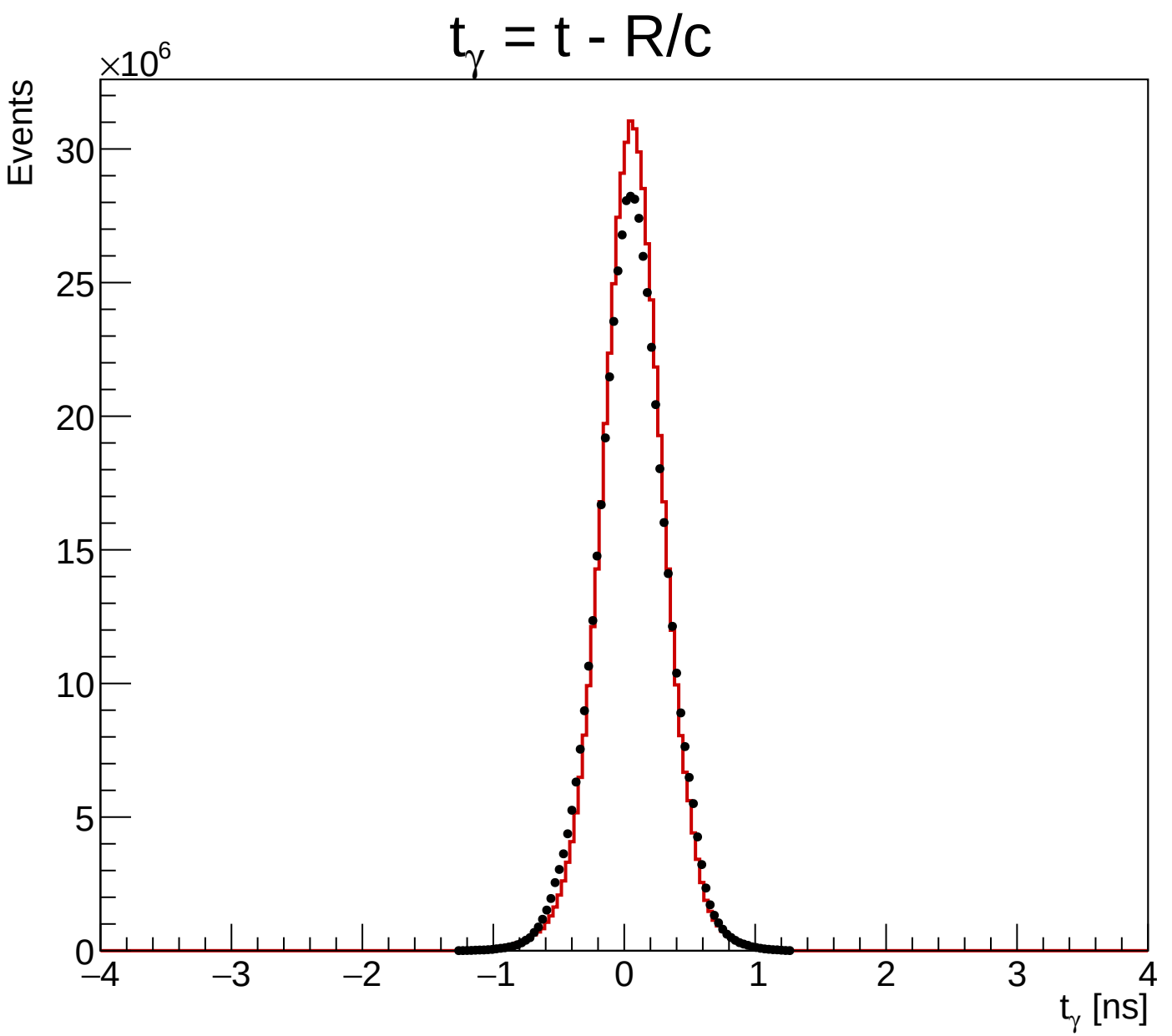

Delta P_y
$\times 10^3$
Events
5000
4000
3000
2000
1000
0
–200
–150
–100
–50
0
50
100
150
200
Δ P_y [MeV/c]

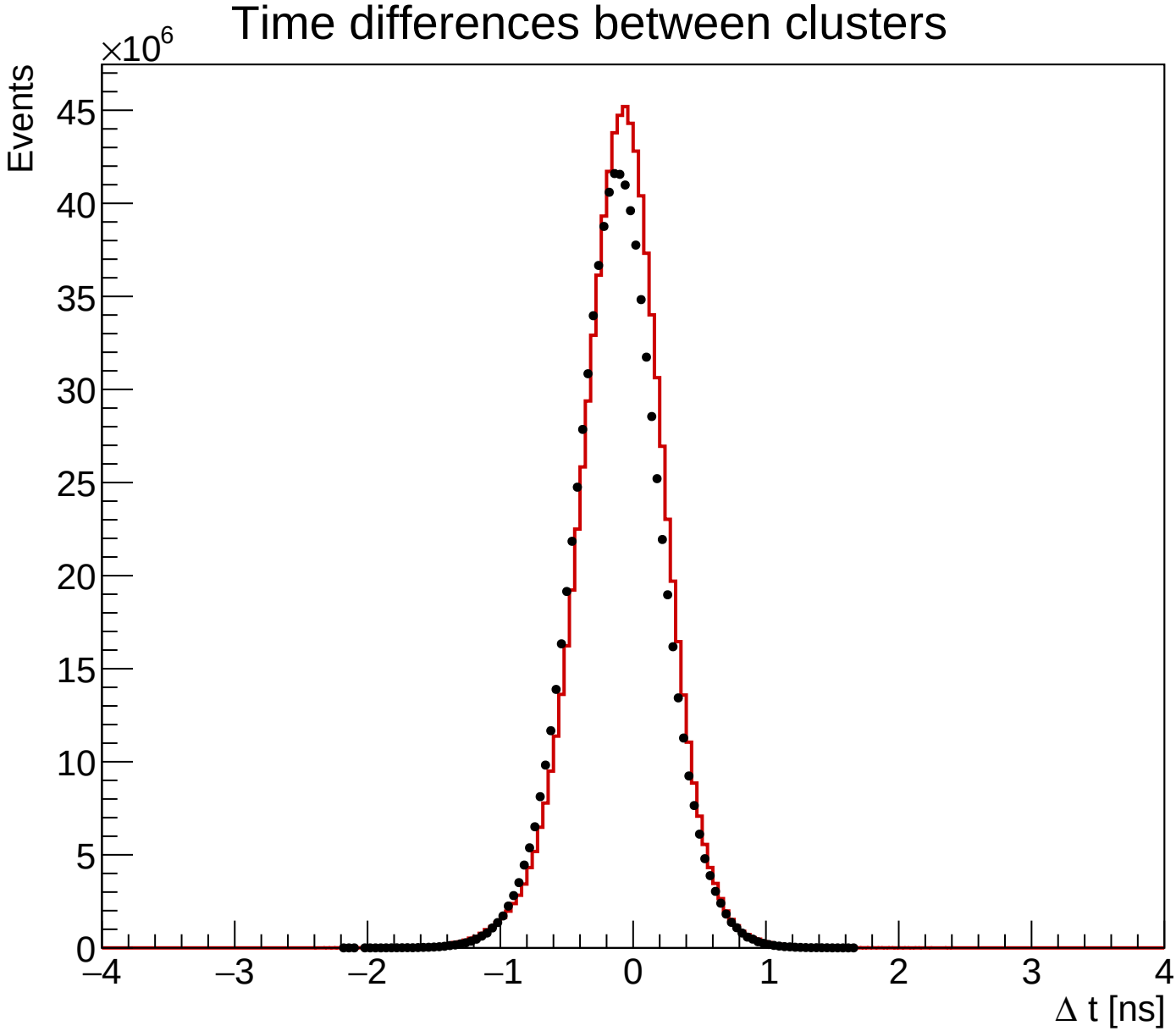

Delta P_z
$\times 10^3$
Events
5000
4000
3000
2000
1000
0
–200
–150
–100
–50
0
50
100
150
200
Δ P_z [MeV/c]

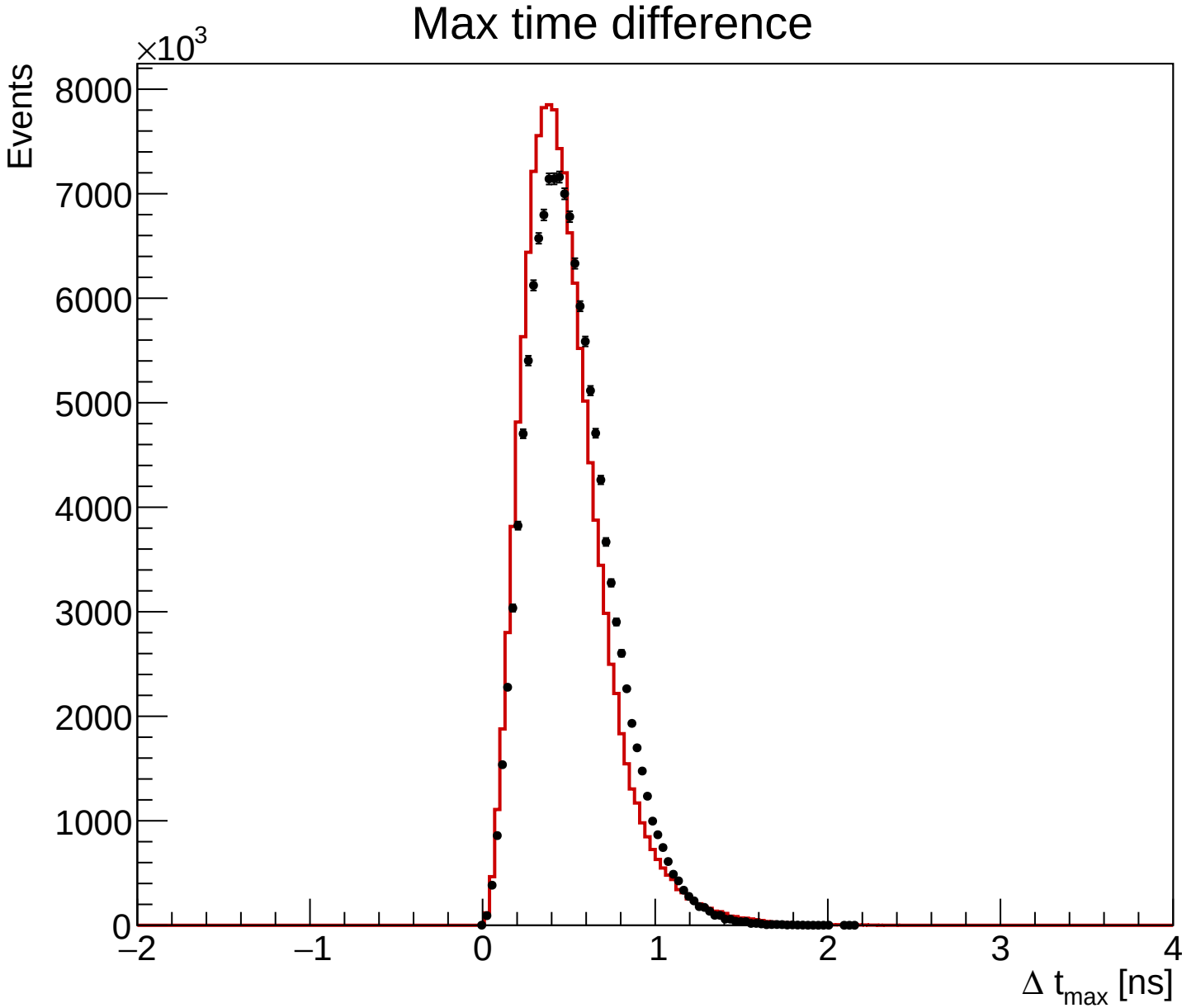

β
×10^3
Events
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
0.18
0.2
0.22
0.24
0.26
0.28
0.3
β

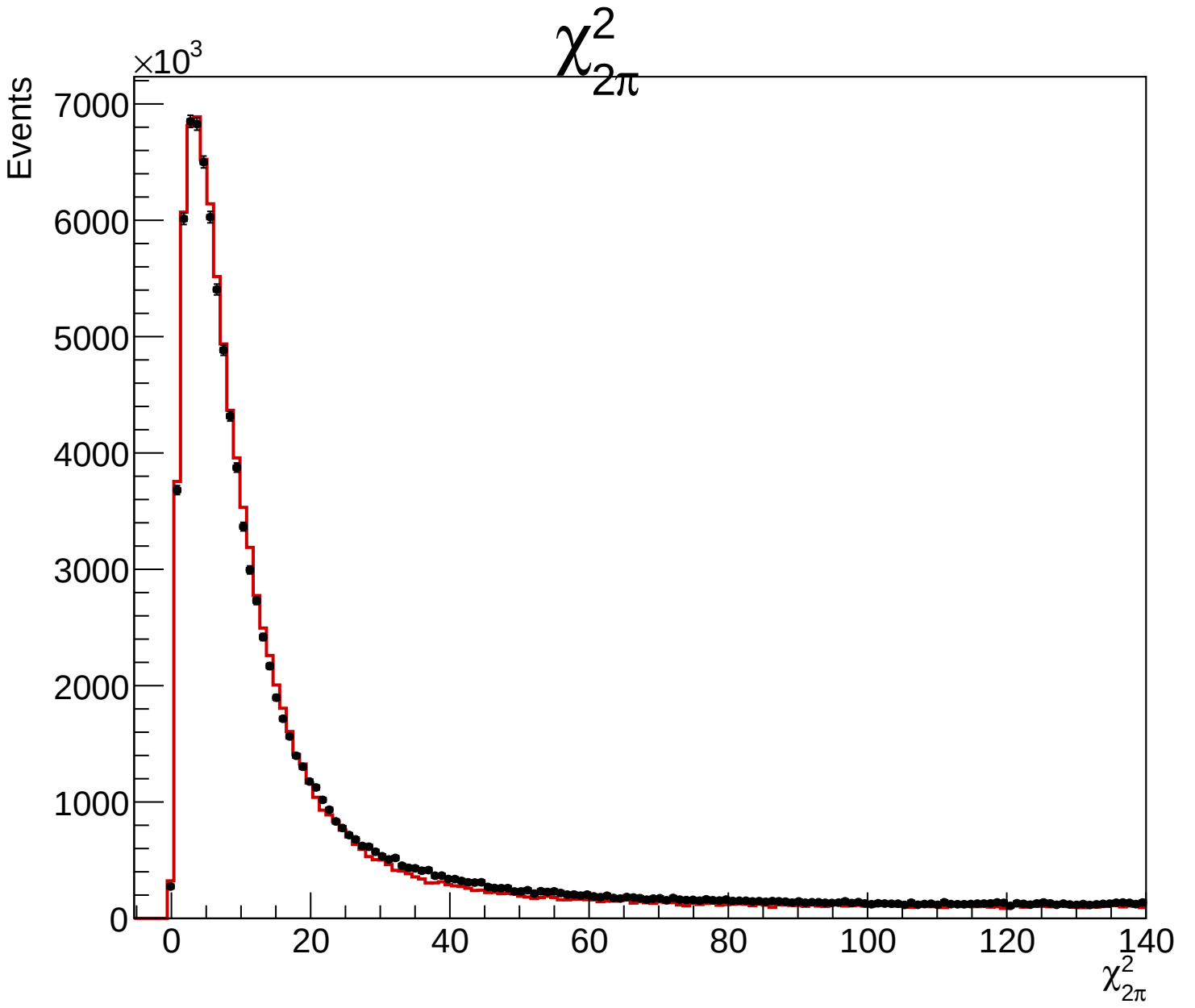

KL crash energy
×10^3
Events
4500
4000
3500
3000
2500
2000
1500
1000
500
0
100
200
300
400
500
600
700
800
E_KL crash [MeV]

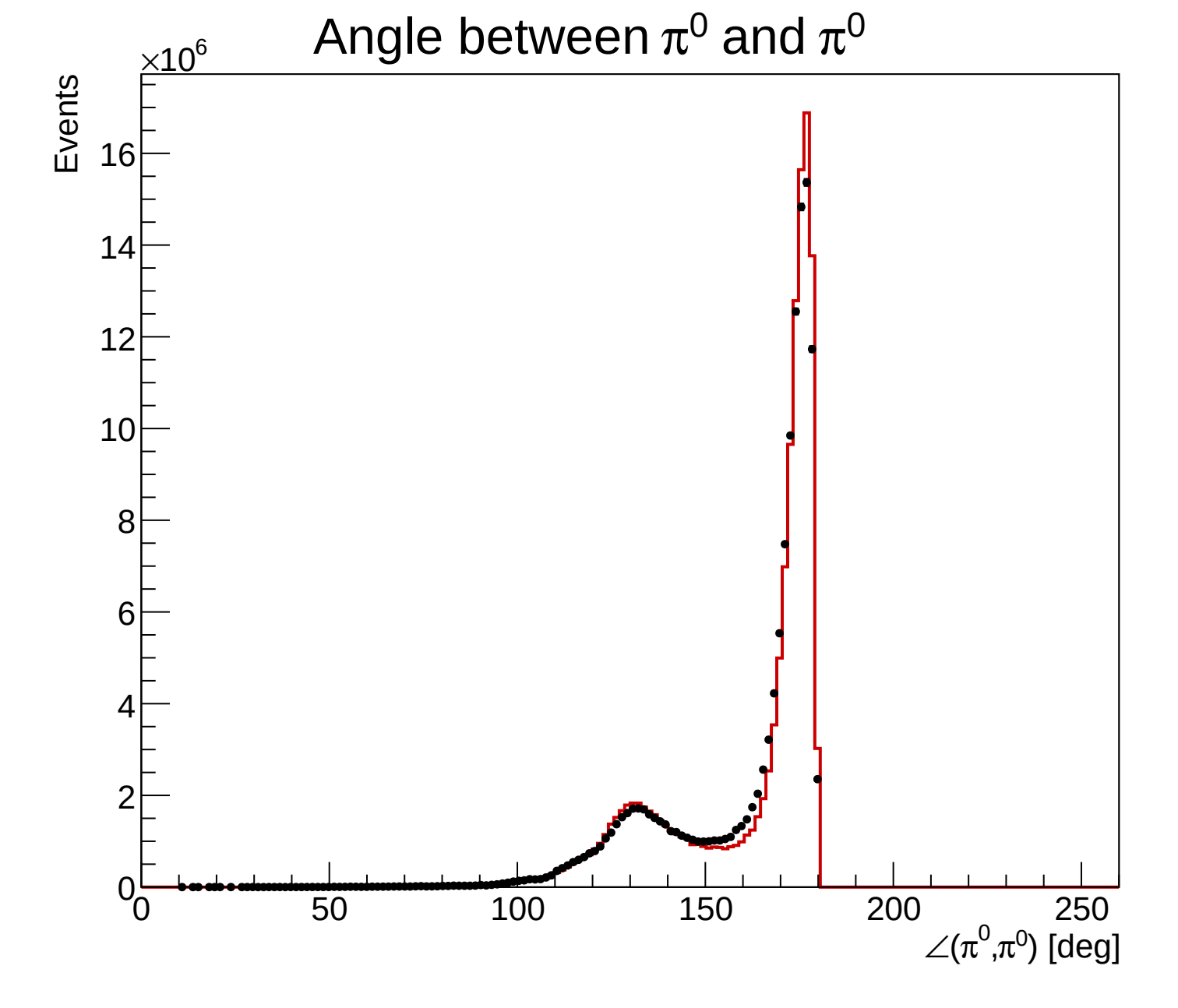

R_{min}
$\times 10^3$
Events
2500
2000
1500
1000
500
0
0
50
100
150
200
250
300
350
400
R_{min} [cm]

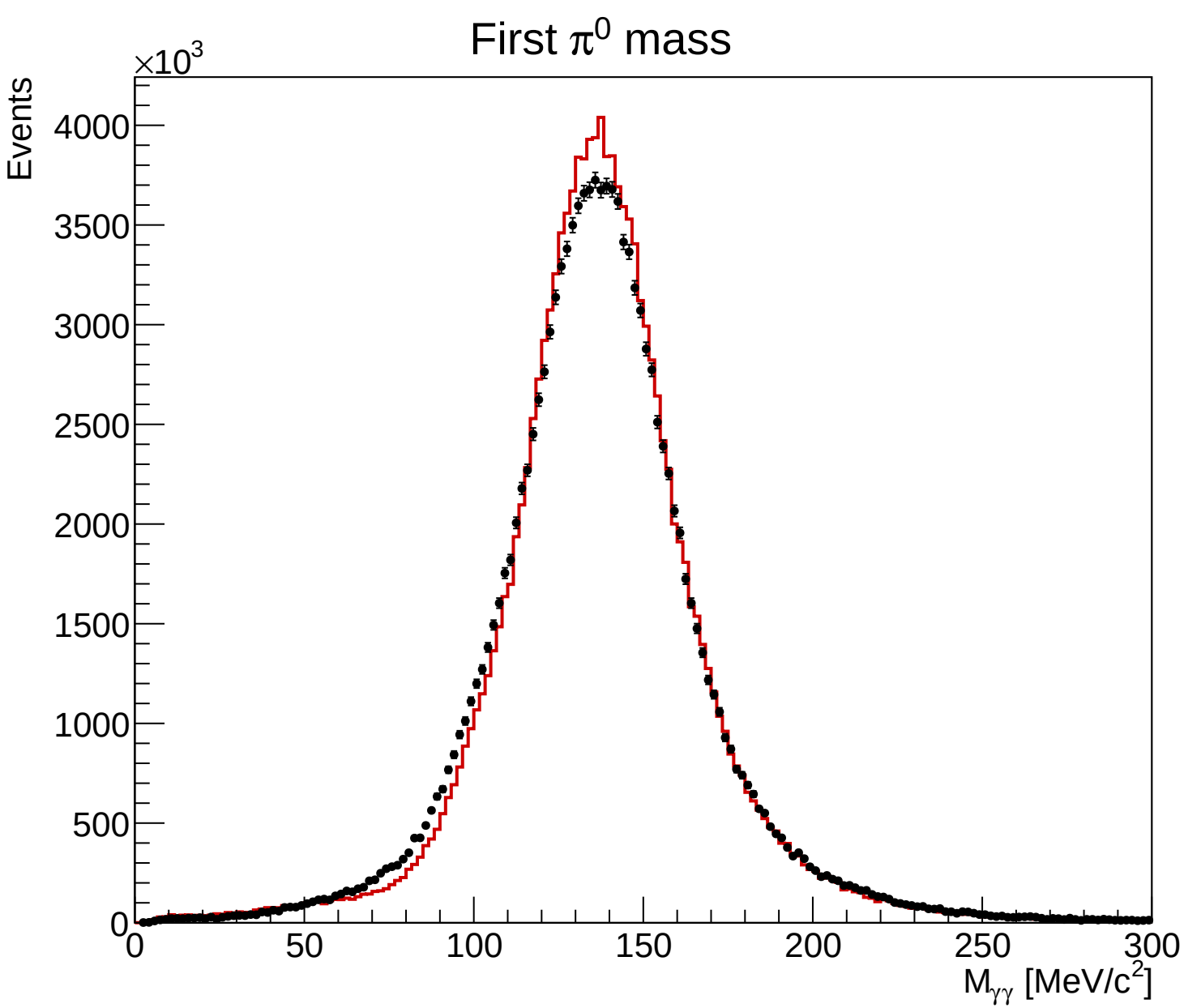

$t_\gamma = t - R/c$
$\times 10^6$
Events
30
25
20
15
10
5
0
−4
−3
−2
−1
0
1
2
3
4
t_γ [ns]

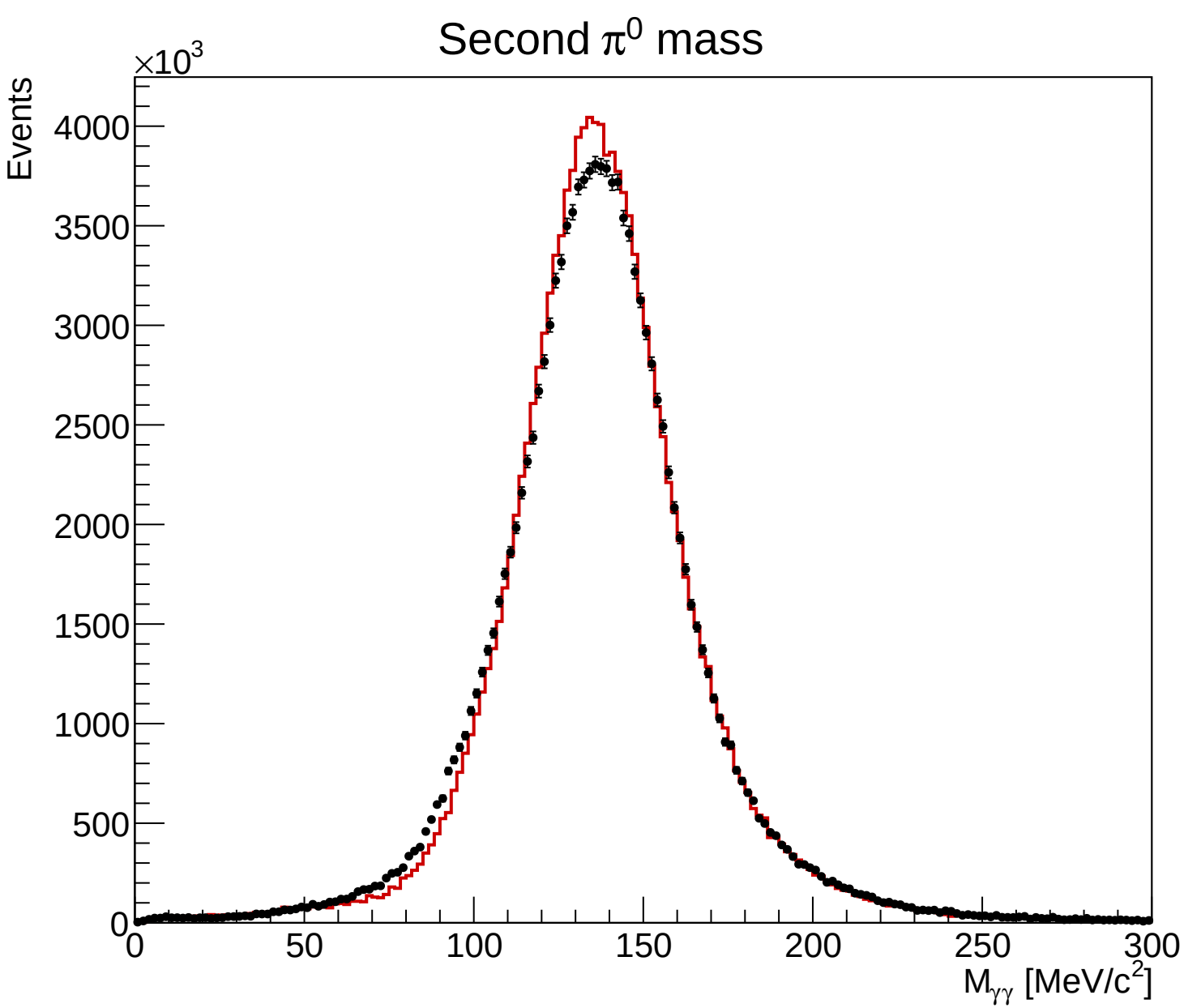

Time differences between clusters
×10^6
Events
45
40
35
30
25
20
15
10
5
0
−4
−3
−2
−1
0
1
2
3
4
Δ t [ns]

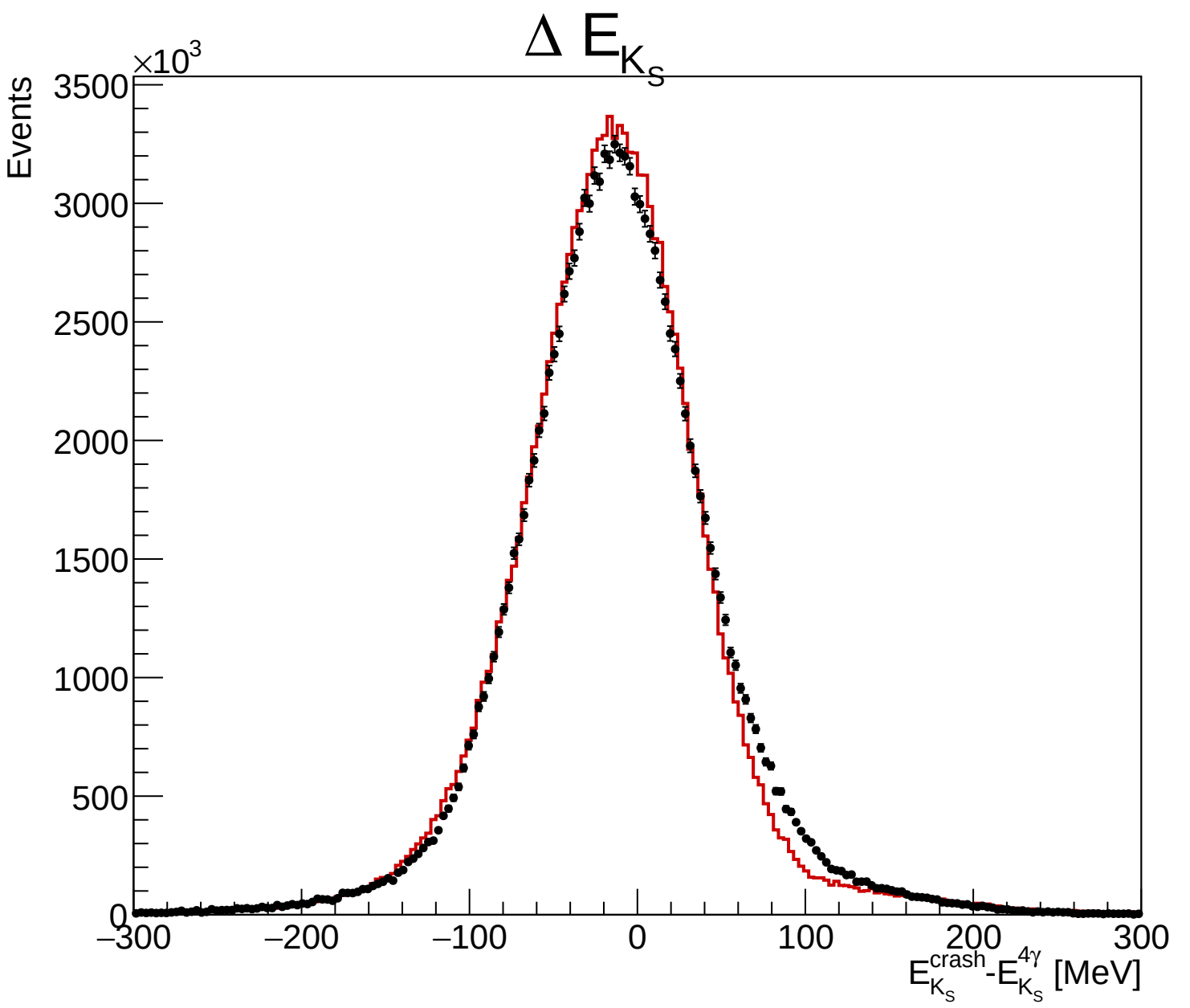

Max time difference
×10^3
Events
8000
7000
6000
5000
4000
3000
2000
1000
0
−2
−1
0
1
2
3
4
Δ t_max [ns]

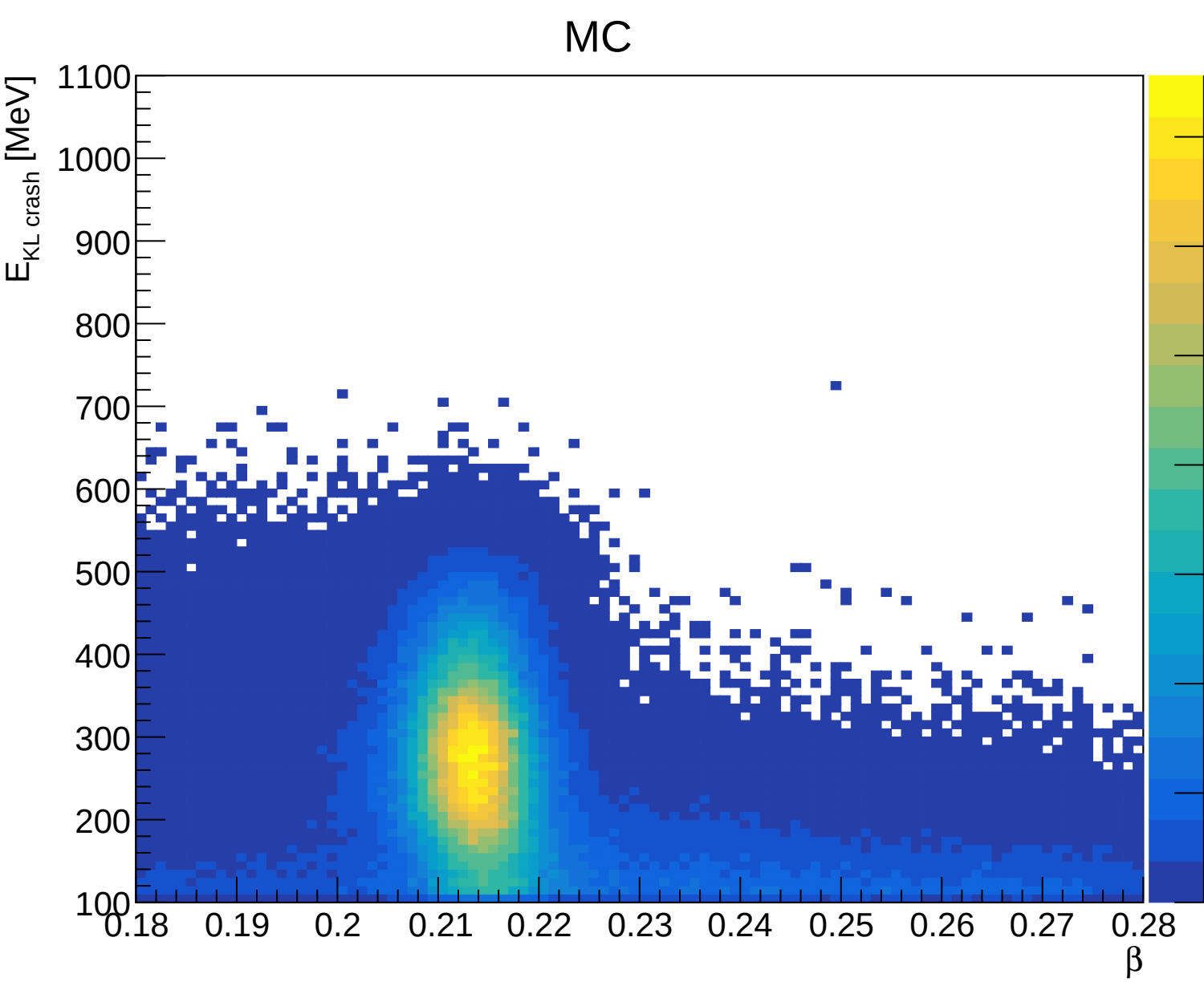

$\chi^2_{2\pi}$
$\times 10^3$
Events
7000
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
$\chi^2_{2\pi}$

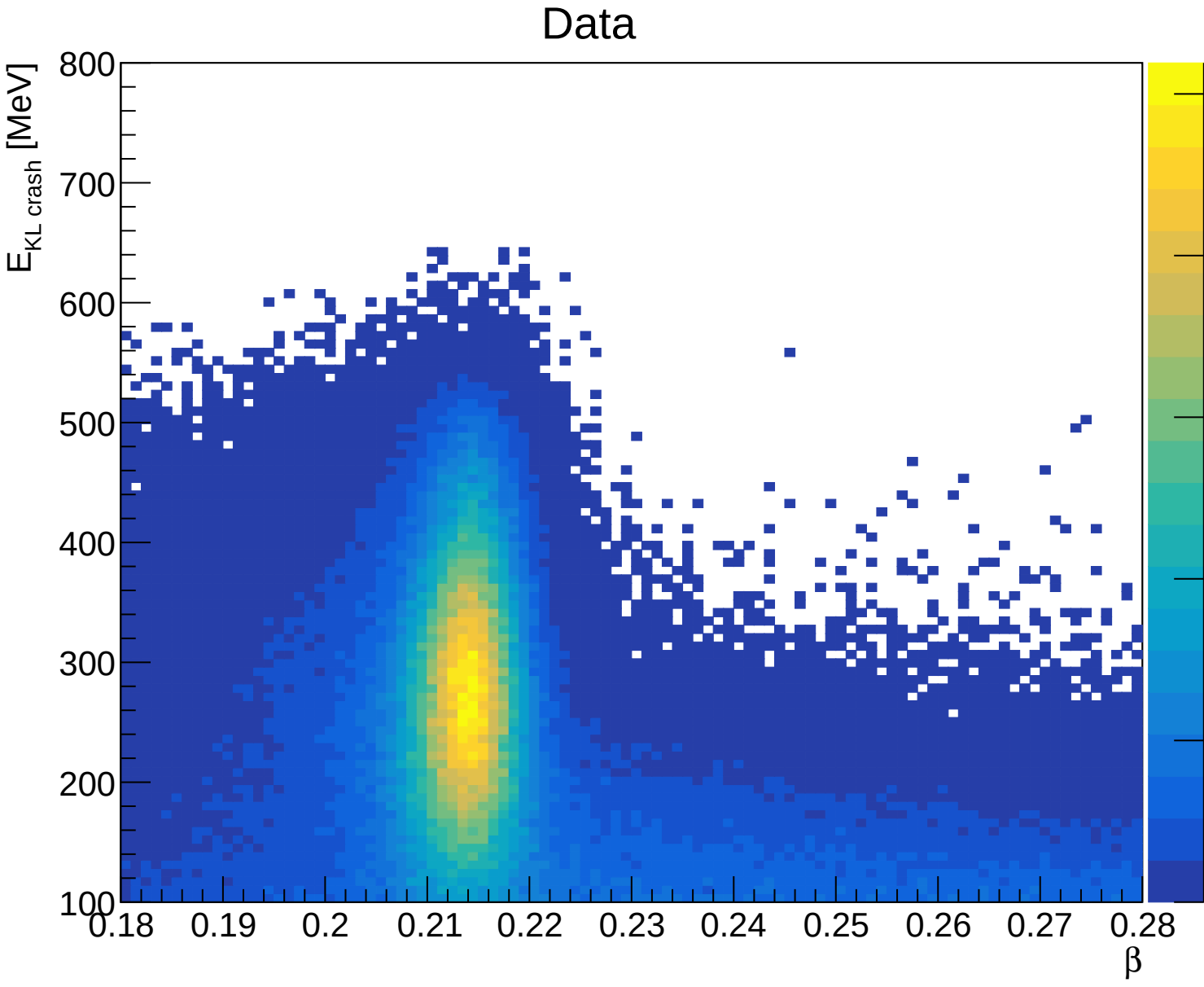

Angle between π^0 and π^0
$\times 10^6$
Events
16
14
12
10
8
6
4
2
0
0
50
100
150
200
250
$\angle(\pi^0,\pi^0)$ [deg]


First π^0 mass
$\times 10^3$
Events
4000
3500
3000
2500
2000
1500
1000
500
0
0
50
100
150
200
250
300
$M_{\gamma\gamma}$ [MeV/c^2]


Second π^0 mass
$\times 10^3$
Events
4000
3500
3000
2500
2000
1500
1000
500
0
0
50
100
150
200
250
300
$M_{\gamma\gamma}$ [MeV/c^2]


Δ E$_{K_S}$
×10^3
Events
3500
3000
2500
2000
1500
1000
500
0
–300
–200
–100
0
100
200
300
E$_{K_S}^{crash}$-E$_{K_S}^{4\gamma}$ [MeV]


MC
$E_{KL\ crash}$ [MeV]
1100
1000
900
800
700
600
500
400
300
200
100
0.18
0.19
0.2
0.21
0.22
0.23
0.24
0.25
0.26
0.27
0.28
β


Data
$E_{KL\ crash}$ [MeV]
800
700
600
500
400
300
200
100
0.18
0.19
0.2
0.21
0.22
0.23
0.24
0.25
0.26
0.27
0.28
β